Cloud Computing for Machine Learning

Introduction to Machine Learning Engineering

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Agenda

- What is Machine Learning Engineering?
- Roles & Responsibilities of an ML Engineer
- ML Engineering vs Data Science
- Machine Learning Development Lifecycle
- Essential Tools & Platforms
- Challenges in ML Engineering
- MLOps
- The Backbone of Effective ML Systems
- Summary
- Q&A

What is Machine Learning Engineering

- Bridging the gap between Data Science and Production
- Focus on scalable ML systems and robust deployment
- Ensuring ML models' reliability and performance

Roles & Responsibilities of an ML Engineer

- Designing ML pipelines
- Scaling and optimizing models for deployment
- Collaborating with data scientists & software engineers
- Monitoring and maintaining ML systems in production

ML Engineering vs Data Science

Data Scientists focus on model creation & validation

ML Engineers focus on model deployment & scaling

Both roles are crucial for an end-to-end ML solution

Machine Learning Development Lifecycle

- Data Collection & Cleaning
- Model Development & Training
- Validation & Testing
- Deployment & Monitoring
- Feedback & Iteration

Essential Tools & Platforms

- TensorFlow & PyTorch for modeling
- Docker for containerization
- Kubernetes for orchestration
- MLflow for ML lifecycle management

Challenges in ML Engineering

- Model drift & versioning
- Scalability concerns
- Data privacy & security
- Continuous integration & delivery for ML

MLOps

Combining ML and DevOps principles

Streamlining deployment and monitoring

Ensuring models are up-to-date and effective

The Backbone of Effective ML Systems

ML Engineering ensures models work in the real world

Collaboration & advanced tools are key

A rapidly evolving field with endless opportunities

Summary

Machine Learning Engineering bridges the gap between Data Science and Production

Roles and Responsibilities of an ML Engineer

ML Engineering vs Data Science: Data Scientists primarily focus on model creation and validation. ML Engineers specialize in model deployment, scaling, and ensuring operational efficacy.

Machine Learning Development Lifecycle: Encompasses stages from Data Collection & Cleaning to Deployment & Monitoring, including Feedback & Iteration for continuous improvement.

Essential Tools: TensorFlow, PyTorch, Dockers, ...

Challenges in ML Engineering: Addressing issues like model drift, scalability, data privacy, and security.

MLOps: The Convergence of ML and DevOps

The Backbone of Effective ML Systems: ML Engineering is crucial for ensuring that models work effectively in the real world.

Q&A